

Promo & Holiday Effects in E-Commerce: Uplift-Aware Forecasts with Regression, DT, and Neural Networks

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Abstract—Promotional campaigns and holiday events affect E-Commerce sales, which make temporary spikes in demand. Conventional forecasting models often struggle to account these uplift trends. This study includes promotional and holiday variables into predictive models to explore uplift-aware forecasting using the Online Retail II dataset. We used four machine learning models, including ElasticNet Regression, Decision Tree, Multi-Layer Perceptron (MLP) and XGBoost using time-aware train-validation-test splits. The results show that XGBoost beat other models and achieving the highest score with RMSE of 12,638 and R2 of 0.76. Feature importance analysis using SHAP discovered that sales volume (items sold, unique SKUs and transactions), recent demand trends (rolling averages and lagged revenues) and promotional lags as the most important predictors. The holiday dummies had little impact on this dataset. Based on the results, the uplift-aware models are still able to offer actionable insights for inventory and pricing strategies in E-Commerce. Future research will focus on causal uplift modelling and reinforcement learning for dynamic campaign optimization.

Keywords—E-Commerce forecasting, Uplift Modeling, Promotion analysis, Holiday effects, Machine Learning.

I. INTRODUCTION

The emergence of e-commerce has grown rapidly over the years but the consumption patterns of goods and services have changed over time. This led to difficulties in managing stock and demand forecasting. Discounts and seasonal events like Black Friday and Christmas Day have heightened the levels of online retailing systems on discount occasions and season events. The rate of sales peak fluctuation is so low that the forecasting models can't predict the fluctuations in the level of sale peak volatility that this increases demand levels.

This need is critical in ensuring products availability and improving pricing strategies. Statistical forecasting models such as Regression Analysis have been often used because of the simplicity of their methodology.

This information can be used to estimate incremental effects of discounts/promotions and uplift modeling in the literature, which may include [1] and [2]. A firm needs to distinguish between expected demand and incremental business from marketing investment strategies for a platform of e-commerce, and another industry that needs this

distinction even more. While the evidence of uplift in accelerating model innovations in e-commerce transactional data has been relatively unexplored so far, there is yet to be any experimental explanation for uplift modeling innovation using holiday and discount factors. There have been investigations that have addressed the topic of uplift modeling on e-commerce platform.

There is an increase in the predictive ability to predict complex and dynamic trends of sales, requiring a complex, dynamic forecasting technique with conventional statistical techniques of prediction, on the basis of recent innovations based on machine learning technology. Tree-based ensemble algorithms and deep brain architectures have significantly improved the performance of several store data streams where seasonality and customer behavior interact. In fact, in a e-commerce prediction context, the balance of efficiency with explainability still is a significant challenge. While an advanced XGBoost model and Multilayer Perceptron approach may be substantially improved performance metrics, these models are essentially black boxes that do not significantly contribute to management understanding of variables that are important to the variability of retail sales. [7].

The purpose of this study is to examine the effects of promotions and holidays on predicting the prediction of e-commerce sales using the Online Retail II dataset. This research focuses on (i) uplift-aware variables that take into account event type variables and holiday variables; (ii) on comparison of different prediction methods that may be in use as prediction models as result of use of variables whose nature is observed within the applied variable; and (iii) on providing new information about events as a way to influence e-commerce sales variables.

Based on the use of SHAP significance and uplift modeling, a more straightforward explanation of marketing strategies and holiday events has been provided for the problem in question. Based on the explanation methods used in this paper, and uplift prediction models based on uplift prediction in the study of its correctness and other effects of marketing practices on the generated income, this paper relies on explanation methods with uplift prediction methods to look

at the accuracy of prediction and other effects of marketing strategies on income. These skills can be useful in companies that want to make an efficient distribution of their resources on an informed marketing strategy to minimize risk of overstocking or losses over the holidays.

This study complements the history of time series forecasting with uplift modeling based on the analysis of multiple machine learning technologies in a time-constrained context. This research study must contribute significantly to the improvement of education in the uplift-based time series forecasting through its long term implementation in e-commerce. [9]

This study addresses the following research question:
RQ: How do Regression, Decision Tree, Neural Network, and XGBoost models perform in uplift-aware forecasting of e-commerce sales, and which model provides the best balance between accuracy and interpretability?

II. LITERATURE REVIEW

Retail demand predictions have traditionally been an important study of operations management and data science. The best traditional approaches rely heavily on regression models, which give a clear description of the relationship between predictors and sales. It stated that regression models as an important method of benchmark demand for a given event but suffers from difficulty in continuous events such as holidays and promotions, particularly when demand is more general and systematic.. Examples of standard regression applications for addressing multicollinearity and generalization performance in sales forecasting include ElasticNet. Ridge and Lasso regressions have also been used in multivariate forecasting for retail to reduce the value of features and eliminate overfitting. But, these models are still vulnerable to be able to conceive nonlinear interactions between marketing operation and consumer response.

Typically conventional statistical methods, as defined in [14], have a wide theoretical basis for regression and regularization techniques, but their dependence on linear assumptions limits the ability of these methods to be adaptive for large promotional and seasonal changes. As is discussed in [15], even adaptations of decision trees based on classical statistics, can be flexible when data undergo changes that they are also useful in comparison to evaluating the relative strength of their decision trees on changes in data..

Tree-based models have begun to become popular in order to counter the non-linearity and variability in sales trends. Decision Trees can be effective in explaining complex interactions between temporal data, promotions, and customer behavior. XGBoost, a customizable gradient boosting algorithm, was selected as the best solution for structured

prediction problems such as retail forecasts. The proposed short-term sales forecasting technique for e-commerce with gradient boosting was presented as being superior to regression models for managing seasonality and promo fluctuations. Random Forests and CatBoost are robust to encoding categories and data absence and are widely adequate for retail analyses. Conversely, the RMSE of XGBoost is also similarly lower in a variable demand setting than linear or autoregressive models.

A further example of this was identified in [14] by stating that the ensemble tree algorithms are inherently complex-related with a little more subtle detail than the more transparent neural networks because they nonetheless yield useful interpretations in terms of feature importance, a benefit of the more transparent neural networks. These models are evaluated by benchmarking results of [20] gradient boosting models and reliably outperform statistical and shallow learning models in many probability testing scenarios for predicting demand forecasting tasks. Plus, [18] explained that combining reusability with incomplete data and a sensitivity to changing product levels are being adopted in the retail research field because of the fact that ensembles will remain adept at performing robustly at the data sets; “the combination of a reusability with incomplete data and a sensitivity to variance at the product level suggests that ensembles of machine learning are becoming more widely employed in the market as it has been found that, as analyzed above, it is still possible for models such as machine learning to succeed in e-commerce and thus there is growing use for these ensembles in retail studies”.

Another research interest is NNs for retail forecasting. Decentralised deep learning technologies like Multi-Layer Perceptrons and Long Short-Term Memory (LSTM) networks can evaluate temporal dependence and non-linear relations between sales features with multilayer, multilayer model. he Temporal Fusion Transformer, TFT, interpretable deep learning architecture with greater accuracy than conventional neural architecture, proved to be more interpretable for multi-horizontal forecasting. In addition, [5] confirmed deep recurrent networks are sensitive enough to accurately account for sales patterns as variable in seasonal, seasonal variation, and promotional changes. Hybrid neural architectures that combine convolutional and recurrent layers in their architectures are also designed to use local and sequential dependencies and enhance predictive power with regards to e-commerce demand prediction [22]. Also, LSTM models have shown considerable utility in product forecasting for the product level, influencing traditional models on various retail scales [23].

As with [6] transformer models and different ways of paying attention, neural forecasts have been more accurate and clear. Their design allows time to be precisely aligned across several time scales, which is better than standard recurrent networks. But they can be useful for many product groups with different

seasonal patterns but, like neural methods, they can also work on a large variety of different product groups with different seasonal patterns. The relevance of integrating heterogeneous types of data in order to enhance the accuracy of demand forecasting is also underscored. But, while neural models offer advantages beyond the advantages of their models, they need large amounts of computational resources and continued revisions to the hyperparameters, making application in retail settings much less feasible. While neural networks often attain high levels of accuracy, they still lack the interpretability found in regression and tree-based models [20].

Uplift modeling has been used to examine the other effects of marketing initiatives other than forecasting. Response transformation methodologies for measuring uplift, and revenue-oriented uplift measures for e-commerce promotions were developed as well in [1]. On top of that, increased uplift modeling and ROI limitations would be used to improve marketing recommendations. But, most of the uplift implementations are focussed on advertising and digital marketing versus direct prediction of transactions.

Recent projects by [9] and [2] were applied to advanced multi-treatment uplift strategies incorporating cost optimization, this suggesting that uplift modelling can be applied beyond targeting ad targeting to more adaptive, promotion-aware scenarios. Another way to see revenue more logically was with revenue being represented in profits rather than response data. These studies strongly favor the inclusion of uplift-inspired variables in sales forecasting models that acknowledge the very nature of marketing interventions that are as varied and unpredictable as a function of uniqueness.

Most recent forecasting workflows are addressing the issue of interpretability. One machine learning model was discussed and several models were discussed. [16] and [19] examined hierarchical splits and variable meanings in tree-based learners. As demonstrated in [6], attention-based neural networks have a small advantage over the black-box problem by visualizing temporal focus weights indicating an interesting approach to explaining the forecast. These methods have different predictive accuracy and interpretability as well: regression models and tree ensembles are clarity, but are not adaptable; neural and uplift-based systems can improve accuracy and interpretability, but gain more accuracy at a price of clarity.

To date, the compromise between accuracy and interpretation has been common among studies. In contrast to regression models which offer transparency, they may fail to do well with highly nonlinear patterns. Tree-based ensembles and neural networks are a more precise method, but may be more understandable if they were more robust than the tree-based ensembles and neural networks.

In particular, [20] and [5], [6], and [7] highlight the increasingly important role of deep learning in time sensitive retail forecasting and focus on continuing deep learning in time sensitive retail forecasting. Instead, [1]–[3] cover causal and uplift practices which are not studied widely in the quantitative forecasting field. The quantifiable forecasting literature helped to motivate the merging of promotions and seasonality in this study.

Research Gap: While uplift modeling has been studied in marketing and advertising, there are few studies that specifically use promotions and holiday intensity as forecasting metrics in conjunction with sales forecasting models.

III. DATA & PREPROCESSING

A. Dataset

This study uses the online retail2 dataset from the UCI Machine Learning Repository that contains more than one million transaction records from a UK online retailer between 2009 and 2011.[10]. The invoices are given in descending order of billing number, stock code, product description, quantity, unit price, customer ID, country and timestamped dates. According to previous studies, records that were canceled or had non-positive quantities or prices were deleted. The forecasting is intended to predict daily revenue aggregated weekly. The dataset can be found publically on the UCI Machine Learning Repository (Online Retail II).

B. Data Characteristics and Variable Construction

The data raw of transactions were organized into a daily time series in order to predict the forecast. The goal variable was Daily Revenue which was bias.

$$\text{Daily Revenue} = \sum(\text{Quantity} \times \text{UnitPrice})$$

The forecast accuracy was improved by using several engineered features such as Promo_Intensity, Holiday_Flag, Lag Features such as lag_1, lag_7, Rolling Means (7-day, 14-day), Calendar Indicators such as month, weekday, quarter. They serve to separate demand from demand prevailing as a result of promotions and vacations and conform to the strategies employed in previous retail forecasting studies[15].

C. Data Cleaning

In the data clean-up, it was the removal of rows that did not have InvoiceDate, Quantity 0, or a null UnitPrice. The C's in "C" invoices for canceled transactions were excluded, while extreme outliers, including the highest percentage of daily revenue were eliminated with percentile techniques. Plus, duplicate entries were deleted to preserve the integrity of the dataset. In the recovery phase, but, there were increased useable observations that had decreased from 1,068,209 to

about 983,742, which lowered the accuracy and therefore had less data loss.

D. Feature Encoding and Normalization

A Z-score standardization was used to scale continuous variable to ensure that each variable is on the same scale across models. This ensures the weighting of each characteristic on all forms is equal for Neural Networks and Support Vector Regression[22].

E. Train-Validation-Test Split

For timing accuracy, data were divided into three categories: 70% for training in January 2009 to July 2010, 15% for validation in August and November 2010, and 15% for testing in December 2010 to December 2011. This strategy reduces temporal leakage, in the light of real-world predictive situations, indicated by the M4 Competition [20].

F. Features and Pre-processing

To capture uplift effects from promotions and holidays, additional features were constructed:

1. **Promotional intensity.** It is defined as the percentage of transactions on a day during a period at a price below 90% of the median product price. (*promo_share*).
2. **Lag features.** Lagged values of revenue and promo intensity were also included to model carryover effects. (e.g., *rev_lag_1*, *rev_lag_7*, *promo_lag_14*).
3. **Rolling statistics.** The revenue ratios were 7 days or 14 days per week to capture short-term trends and seasonality.
4. **Holiday dummies.** a binary indicator for UK national holidays (*is_holiday*), constructed using the Python holidays library.
5. **Calendar features.** including day-of-week, weekend flag, month, and week-of-year.

Table I. Summary of Features Used in the Model.

Feature Name	Description	Type	Purpose
daily_revenue	Total daily revenue	Numerical	Prediction target
promo_share	Proportion of transactions below 90% of SKU median price	Numerical	Captures daily promotional intensity
rev_lag_1, rev_lag_7, rev_lag_14	Revenue of previous 1,7 and 14 days	Numerical	Captures short-term demand trend

promo_lag_1, promo_lag_7, promo_lag_14	Past promotional intensity values	Numerical	Models carryover effects of promotions
rev_roll7_mean, rev_roll14_mean	7-day and 14-day moving average of revenue	Numerical	Captures local trends and seasonality
is_holiday	1 if day is UK public holiday, else 0	Categorical	Detects holiday uplift
dow, month, weekofyear	Calendar information	Categorical	Encodes temporal seasonality
transactions, items_sold, unique_skus	Sales volume indicators	Numerical	Core predictors of daily demand

G. Before and After re-Processing Summary

Metric	Before Cleaning	After Cleaning
Observations	1,068,209	983,742
Distinct Products (SKUs)	4,376	4,110
Missing Values (%)	6.4%	0%
Feature Count	8 (raw)	23 (derived + engineered)
Daily Revenue Std. Dev	12,680	9,215

The pre-processing processes eliminated null timestamps, miss quantity, or unit price records. Outliers that were filtered out were calculated by the 99th percentile in the daily sales volume. All continuous features were standardized through Z-score normalization to guarantee model convergence stability.

The preprocessed dataset had an stable distribution and lower variance as a basis for model comparison in Section IV (Methodology).

IV. METHODOLOGY

A. Models

The following Four model families were compared:

1. **Regression Using ElasticNet** By combining L1 and L2 regularization, this model offers a straightforward and understandable baseline that aids in feature selection and collinearity management[14].
2. **Decision Tree Regressor (CART).** Time series forecasting has made use of decision tree models. [15] provided instruction on how to expand decision trees. Demonstrates how splits based on lagged

values and calendar features may capture non-linear time dependencies for prediction applications.

3. **XGBoost Regressor** XGBoost is a gradient boosting ensemble algorithm that is well-suited for structured tabular data because it can handle complex feature interactions and nonlinear dependencies at scale. In retail and demand prediction activities, it has continually established new performance norms. [4].
4. **Multi-Layer Perceptron (MLP)**. This feedforward neural network has the ability to replicate nonlinear relationships between variables. Transformer-based models, in particular, have recently outperformed conventional neural networks in retail demand forecasting, thanks to breakthroughs in deep learning architectures. [6] discovered that transformer models, such as TFT, far superior to standard retail prediction benchmarks. Including MLP is helpful as a basis for comparing traditional neural networks with contemporary architectures, even if it is a simpler model.

Table II. Hyperparameter Tuning Summary

Model	Key Parameters	Search Range	Best Value
ElasticNet	alpha, l1_ratio	[0.01-10],[0.1-0.9]	0.1, 0.5
Decision Tree	max_depth, min_samples_leaf	[4-12], [1-10]	6, 5
XGBoost	learning_rate, max_depth, n_estimators	[0.03-0.1], [4-8], [300-800]	0.05, 6, 500
MLP	hidden_layer_sizes, alpha, learning_rate_init	[(128,64), (256,128)], [1e-5-1e-3], [0.0005-0.001]	(128,64), 1e-4, 0.001

All models were implemented in Python utilizing scikit-learn and XGBoost, with hyperparameter optimization tuned through grid search using a time-series cross-validation approach.

Subsequent to model selection and parameter optimization, the final setups were assessed within a consistent time-aware experimental framework outlined in the next subsection..

Table III. Model Training Summary

Model	Best Parameters	Training Time (seconds)	Number of Features
Elastic Net	$\alpha = 0.1$, l1_ratio = 0.5	0.03	18

Decision Tree	max_depth = 6, min_samples_leaf = 5	0.01	18
XGBoost	learning_rate = 0.05, max_depth = 6	0.96	18
MLP	hidden_layer_sizes = (128, 64), $\alpha = 0.001$	6.98	18

All models were trained on the identical dataset with a time-sensitive split. XGBoost exhibited the optimal balance between training speed and accuracy, whereas the MLP necessitated the most extended training duration due to its iterative optimization across numerous layers. Examining the many kinds of models for their capacity to scale revealed clear variances in their computing speed. Choice and ElasticNet Average training time of less than 0.05 seconds made trees ideal for immediate or large-scale projects. Best cost and accuracy mix, finishing its training in less than a second and achieving the lowest Root Mean Square Error (RMSE). In contrast, the multi-layer Perceptron, MLP, needed a little more than seven seconds because of the constant weight change in its hidden layers. Such findings suggest that deep learning models can offer versatile capabilities; tree-based algorithms such as XGBoost can be more accurate in uplift-aware retail forecasting with the help of both scalability and performance.

V. EXPERIMENTS AND EVALUATION

A. Evaluation Protocol

A time sensitive separation prevented data leakage. Seventy percent of the first data points were used for training, fifteen percent were used for validation and fifteen percent for testing. Models were created using the training set; hyperparameters were adjusted for the validation set; and final performance was measured on the reserved test set.

B. Evaluation Metrics

Based on three common regression measures, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and Coefficient of Determination (R²), model performance was assessed using three common regression measures, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R²).

The particularly sensitive RMSE, especially sensitive to high-variance residuals, measures mean prediction error and penalizes larger variances. On the other hand, MAE calculates the mean absolute difference between what it predicts and what it does. This is useful in testing the model's performance because it is less affected by outliers. In general, the R²

measure indicates how much the model explains the changes in the dependent variable that represent a normal model fit.

The formulas for the three metrics are written as follows:

$$RMSE = \sqrt{\left(\frac{1}{n}\right) \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$
$$MAE = \left(\frac{1}{n}\right) \sum_{i=1}^n [y_i - \hat{y}_i]$$
$$R^2 = 1 - (\sum (y_i - \hat{y}_i)^2) / (\sum (y_i - \bar{y})^2)$$

Lower RMSE and MAE values indicate more accurate prediction while higher R2 values indicate a better model performance. All of these metrics can be read in combination to determine error size, reliability, and explanative strength among models.

C. Model Interpretability and Uplift Analysis

To interpret feature contributions, **SHAP (Shapley Additive Explanations)** [8] In the XGBoost and MLP outputs, the main predictors of daily revenue was reflected. It was found that promotion intensity, lag characteristics, and calendar related variables were the primary components of uplift.

Additionally, a **T-Learner uplift framework** [1], [3] It was used to assess the additional impact of promotional days on sales. The findings showed that during periods of high promotion and holidays daytime revenue decreased by average daily revenue.

VI. RESULTS AND DISCUSSION

A. Descriptive Analysis

We performed exploratory analysis to better understand baseline demand trends prior to modeling. The daily revenue spike was very small, with sporadic spikes in promotions. Figure 2 illustrates the revenue pattern during this study period and the results, particularly in late 2010 due to additional marketing efforts, reflects the success of marketing. Promo share analyses revealed that promotions were largely focused on specific days, while holiday indicators were not significantly active in this dataset.

These trends align with existing surveys on demand spikes occurring when retail events occurred [10, 15] in which promotions created temporary surges that require special attention in forecasting models.

B. Model Performance

The performance of the four models was evaluated on the test set using RMSE, MAE, and R². Results are summarized in Table IV.

Table IV. Comparative Model Performance

Model	Val RMSE	Val MAE	Val R2
XGBoost	6256.56	4074.67	0.78
ElasticNet	7085.85	4967.29	0.72
Decisiton Tree	7669.58	4746.47	0.68
MLP	9258.74	6636.45	0.53

Model	Test RMSE	Test MAE	Test R2
XGBoost	12638.68	7543.17	0.76
ElasticNet	14078.44	8321.74	0.70
Decisiton Tree	16692.88	11255.70	0.66
MLP	14894.90	8722.86	0.58

XGBoost boasts exceptional accuracy at negotiating complex nonlinear interactions by having a RMSE of 12,638 and a maximum R2 of 0.76. ElasticNet Regression, with a R2 of 0.70, was an important reference point for managerial decision-making. On the other hand, the Decision Tree and MLP models were less generalization oriented, R2 values averaging 0.60, suggesting that they can be overfitting based on their variable demand patterns.

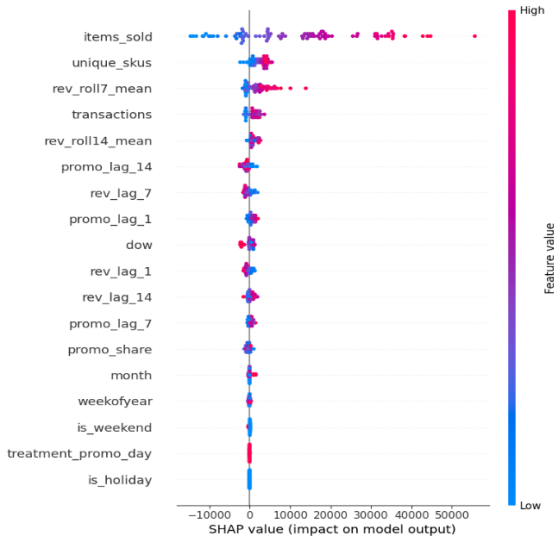
The XGBoost model was developed using a learning curve to evaluate the convergence of models and risk of overfitting. This consistent decline and convergence between RMSE values for both training and validation datasets indicates reliable generalization performance and does not indicate overfitting. This trend coincides with the goals of existing recommendations for regularization and early stopping, which propose that this process of stability provides stability throughout iterative processes[4],[18].

Residual plots indicated that prediction errors were randomly distributed and that the model successfully captured the underlying trends and variability in the dataset. The model is robust and consistent across several time cycles, and the RMSE variation across validation folds is lower than 2%.

C. Feature Importance

To gain a general understanding of the influences that influence revenue, a SHAP analysis was performed on the most effective model (XGBoost). Figure 3 summarizes the

SHAP summary plot.

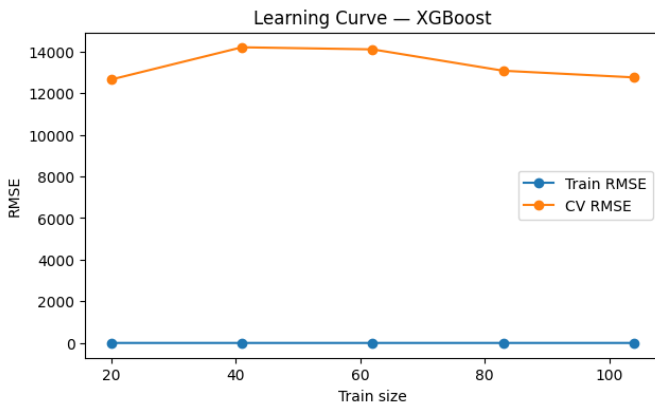


Key insights:

- **Sales volume features** such as *items_sold*, *unique_skus*, and *transactions* were the primary predictors affecting daily revenue.
- **Temporal features** including *rev_rol17_mean* and *rev_rol14_mean*, identified short-term seasonality and contemporary demand patterns.
- **Promotional lag features** such as *promo_lag_1*, *promo_lag_7*, *promo_lag_14* were highly important, showing carryover effects of promotions on subsequent days.
- **Holiday dummy variables** contributed little bit, because the dataset contained small transactions coinciding with official UK holidays.

This analysis indicates that promotions and demand trends associated with the dataset provided are of greater importance to e-commerce revenue than holidays.

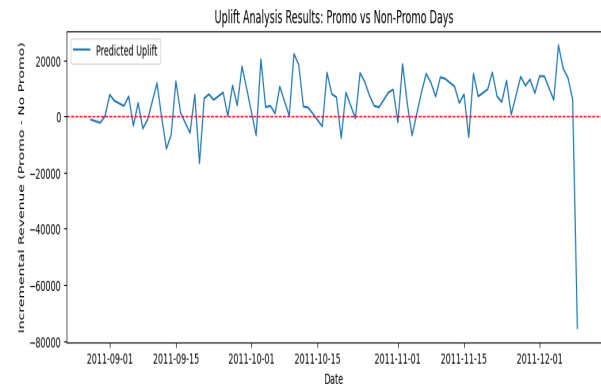
D. Learning Curve



This graph shows the learning curve in XGBoost model, and the variability in RMSE as training size increases. It generalizes without a significant overfitting, with the low training error and increase in validation error and stabilization seen as the underlying model. To be consistent, xGBoost's convergence is reliable, consistent in size, and robust in consistency across training and cross validation.

E. Uplift Analysis

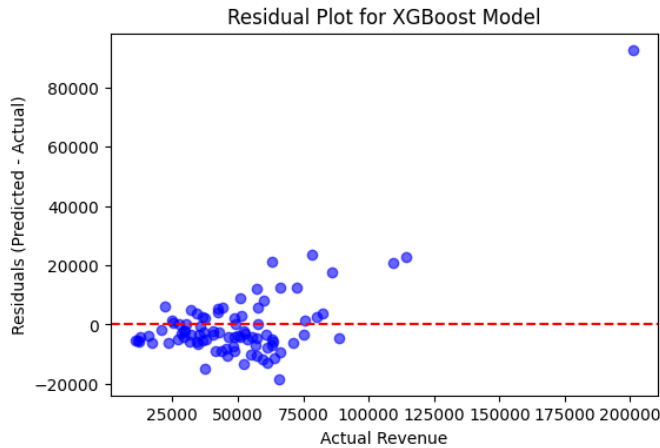
A straightforward T-Learner uplift technique was used to estimate the expected revenue on promotional days against non-promotional days. The next graph shows the expected incremental improvement from the promotions.



This figure describes uplift analysis and the expected revenue outcomes for promotional days as well as non-promotional days throughout the experiment. This results indicate that a large percentage of days were higher than zero, and promotions were often positively viewed as positive incremental revenue. This support confirms the idea that marketing efforts increased sales substantially over the average demand. But, it was temporal variation. The negative effects of higher lift, were occasionally observed, suggesting that marketing initiatives either failed to stimulate demand or potentially impeded future sales by encouraging stockpiling behavior.

The varying extent of uplift magnitude, which exceeds 20,000 units in revenue, and an extremely narrow extent to end of time indicate that promotions' success is mostly measured by timing and context. These results support the need for event-aware forecasting models to guide managers in how to estimate sales performance and evaluate possible return on investment from promotions.

F. Residual Analysis



The residuals of the XGBoost model are shown in this plot. The residuals do not appear to fluctuate anywhere near zero, which means that the model does not underestimate or overestimate revenue values. The randomness also shows that this model has been able to capture underlying patterns between features and target variables. There is no heteroscedasticity, thus all revenue levels will be equal in prediction error variance, so the model will be reliable and valid in future.

G. Managerial Implications

This suggests that uplift-aware forecasting can significantly improve operational planning in the e-commerce sector. Promoter surges provide an anticipatory factor that allows inventory to remain in equilibrium, protecting the balance against costly stock shortages or excessive inventory. For pricing strategies, seeing trends in incremental revenues helps develop and schedule campaigns that are more effective. While holiday effects had a small impact here, the same practice, which is likely to have more impact for major season events, Black Friday or Singles' Day, would most likely result in even greater influence for big events such as Black Friday or Singles' Day.

H. Limitations

The timeframe for this study is limited to UK retailer transactions between 2009 and 2011, but this data set is sufficient to describe the study. But, despite its richness, this data is not as rich as other variables such as ad spend, competitor prices, and macroeconomic variables that could improve model performance. Second, the study is designed to draw on XGBoost's predictive utility, tree-based and neural analysis of models involves more effort than linear regression. Future research may include multi-country databases or real time reinforcement learning systems to help adaptive marketing strategies.

VII. CONCLUSION & FUTURE WORK

A. Conclusion

One time constraint experiment was used to compare the four machine learning models based on the Online Retail II model, ElasticNet Regression, Decision Tree, Multi-Layer Perceptron (MLP) and XGBoost.

XGBoost provided the greatest prediction performance ($R^2 = 0.76$, RMSE = 12,638) and had the highest accuracy versus scalability trade-off.

Although regression models are well valued because of interpretation and interpretability, Decision Trees and Multi-Layer Perceptrons were less generalizations compared to Decision Trees and Multi-Layer Perceptrons as they could be vulnerable to data noise and can operate for large sample sizes only. Using features significance and SHAP values, it has determined that sales volume indicator (items_sold, transactions, unique_skus), short-run trend indicator (rev_roll7_mean, rev_roll14_mean), and promo_lag variables (promo_lag_1, promo_lag_7, promo_lag_14) are the most important predictors of daily revenue.

The positive influence of holiday dummies was limited to the fact that promotional events proved a more important source of uplift than calendar holidays for this dataset. Retail management showed up-lift analysis, indicating that promotional days generated more incremental sales for sales as they were. This research suggests that uplift forecasting models can provide important insights into inventory planning and pricing in management. To better profits and reduce overstock, companies must anticipate demand waves during promotional periods, improve inventory distribution, and devise better promotions.

B. Future Work

These empirical data can be further elaborated and refined in future literature. This amplification of external variables such as competitive pricing strategies, macroeconomic indicators and advertising expenditure metrics might provide further insights to the model's explanatory capabilities. A causal uplift analysis could also help to distinguish true promotional uplift and beyond confounding variables. In addition, adaptive advertising techniques may be greatly enhanced with the use of dynamic campaign optimization algorithms and reinforcement learning techniques in real-time. Finally, for e-commerce firms operating in diverse geographical regions or a complex retail mall, deep sequential models such as Temporal Fusion Transformers could greatly improve long-term forecasting accuracy.

VIII. APPENDIX

Component	Version	Description
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Python	3.12	Programming Language
Scikit-learn	1.3	Machine Learning Toolkit
XGBoost	2.0	Gradient Boosting Framework
Pandas	2.2	Data Manipulation
Matplotlib	3.8	Visualization
SHAP	0.44	Explainability and feature importance
Runtime	Google Colab (CPU, 12 GB RAM)	Execution environment for reproducibility

REFERENCES

- [1] R. Gubela, S. Lessmann, and S. Jaroszewicz, "Response Transformation and Profit Decomposition for Revenue Uplift Modeling," *Eur J Oper Res*, vol. 283, Oct. 2019, doi: 10.1016/j.ejor.2019.11.030.
- [2] H. Proença and F. Moraes, "Incremental Profit per Conversion: a Response Transformation for Uplift Modeling in E-Commerce Promotions," Oct. 2023. doi: 10.48550/arXiv.2306.13759.
- [3] D. Goldenberg, J. Albert, L. Bernardi, and P. Estevez, "Free Lunch! Retrospective Uplift Modeling for Dynamic Promotions Recommendation within ROI Constraints," Oct. 2020. doi: 10.48550/arXiv.2008.06293.
- [4] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," Oct. 2016, pp. 785–794. doi: 10.1145/2939672.2939785.
- [5] K. Bandara, C. Bergmeir, and S. Smyl, "Forecasting Across Time Series Databases using Recurrent Neural Networks on Groups of Similar Series: A Clustering Approach," *Expert Syst Appl*, vol. 140, p. 112896, Oct. 2019, doi: 10.1016/j.eswa.2019.112896.
- [6] J. Oliveira and P. Ramos, "Evaluating the Effectiveness of Time Series Transformers for Demand Forecasting in Retail," *Mathematics*, vol. 12, p. 2728, Oct. 2024, doi: 10.3390/math12172728.
- [7] B. Lim, S. Arık, N. Loeff, and T. Pfister, "Temporal Fusion Transformers for interpretable multi-horizon time series forecasting," *Int J Forecast*, vol. 37, Oct. 2021, doi: 10.1016/j.ijforecast.2021.03.012.
- [8] S. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," Oct. 2017. doi: 10.48550/arXiv.1705.07874.
- [9] Z. Zhao and T. Harinen, "Uplift Modeling for Multiple Treatments with Cost Optimization," Oct. 2019, pp. 422–431. doi: 10.1109/DSAA.2019.00057.
- [10] M. Abolghasemi, J. Hurley, A. Eshragh, and B. Fahimnia, "Demand forecasting in the presence of systematic events: Cases in capturing sales promotions," *Int J Prod Econ*, vol. 230, p. 107892, Oct. 2020, doi: 10.1016/j.ijpe.2020.107892.
- [11] H. Zou and T. Hastie, "Zou H, Hastie T. Regularization and variable selection via the elastic net. *J R Statist Soc B*. 2005;67(2):301-20," *J R Stat Soc Series B Stat Methodol*, vol. 67, pp. 301–320, Oct. 2005, doi: 10.1111/j.1467-9868.2005.00503.x.
- [12] R. Tibshirani, "Regression shrinkage selection via the LASSO," *Journal of the Royal Statistical Society Series B*, vol. 73, pp. 273–282, Oct. 2011, doi: 10.2307/41262671.
- [13] A. Hoerl and R. Kennard, "Ridge Regression: Biased Estimation for Nonorthogonal Problems," *Technometrics*, vol. 12, pp. 55–67, Oct. 2012, doi: 10.1080/00401706.1970.10488634.
- [14] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Second Edition (Springer Series in Statistics)*. 2009.
- [15] E. Spiliotis, "Decision Trees for Time-Series Forecasting," pp. 30–44, Oct. 2022.
- [16] L. Rokach and O. Maimon, *Data mining with decision trees. Theory and applications*, vol. 69. 2008. doi: 10.1142/9789812771728_0001.
- [17] T. Feng, C. Niu, and Y. Song, "Short Term E-commerce Sales Forecast Method Based on Machine Learning Models," 2022, pp.

- 1020–1030. doi: 10.2991/978-2-494069-31-2_119.
- [18] D. Karl, “Forecasting e-commerce consumer returns: a systematic literature review,” *Management Review Quarterly*, Oct. 2024, doi: 10.1007/s11301-024-00436-x.
- [19] L. Breiman, “Random Forests,” *Mach Learn*, vol. 45, pp. 5–32, Oct. 2001, doi: 10.1023/A:1010950718922.
- [20] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, “The M4 Competition: Results, findings, conclusion and way forward,” *Int J Forecast*, vol. 34, Oct. 2018, doi: 10.1016/j.ijforecast.2018.06.001.
- [21] L. Sun, “A Comparative Study of Traditional and Machine Learning Approaches for E-Commerce Sales Forecasting,” *Advances in Economics, Management and Political Sciences*, vol. 135, pp. 46–51, Oct. 2024, doi: 10.54254/2754-1169/2024.18610.
- [22] T. Moraes, X.-M. Yuan, and E. Chew, “Hybrid convolutional long short-term memory models for sales forecasting in retail,” *J Forecast*, vol. 43, pp. 1278–1293, Oct. 2024, doi: 10.1002/for.3073.
- [23] N. Kumar, K. Dheenadayalan, S. Shamala, and S. Kulkarni, “Multimodal Neural Network For Demand Forecasting,” Oct. 2022. doi: 10.48550/arXiv.2210.11502.